

Aero Intelligence: Predictive Machine Learning for Active Aerodynamics Efficiency and Driver Anomaly Analysis in Formula 1

A Proof-of-Concept Pipeline: Case Study of the 2026 Japanese Grand Prix at Suzuka Circuit

Abstract—Modern Formula 1 aerodynamic regulations (2026 era) mandate active systems transitioning between high-downforce Z-Mode and low-drag X-Mode configurations. This paper presents Aero Intelligence, a four-phase ML pipeline applied to high-frequency telemetry from the 2026 Japanese Grand Prix at Suzuka, covering both Red Bull Racing drivers: Max Verstappen (VER) and Isack Hadjar (HAD). The pipeline applies K-Means clustering ($k=4$, Silhouette=0.6743) for track zone identification, Isolation Forest for driver anomaly detection (5.00% rate, $3.5\times$ deviation correlation), Logistic Regression as an interpretable baseline ($F1=0.9092$, $AUC=0.9421$), and Random Forest as the primary model ($F1=0.9801$, $AUC=0.9887$). Physics-based features — Kinetic Energy, Longitudinal Force, and Energy Efficiency Ratio — ranked as the top predictive signals. 5-fold cross-validation and DeLong AUC tests confirm RF significantly outperforms the logistic baseline ($p<0.001$), with results suggesting ML-driven aerodynamic advisory systems can reduce timing errors by up to $5\times$.

Index Terms—Formula 1 telemetry, active aerodynamics, K-Means clustering, Isolation Forest, Random Forest, anomaly detection, machine learning, sports analytics.

I. INTRODUCTION

The 2026 Formula 1 technical regulations introduced a paradigm shift: active aerodynamic systems capable of transitioning between Z-Mode (high downforce) for cornering and X-Mode (low drag) above 240 km/h for straight-line velocity. The critical engineering question is not which mode to use — that is governed by physics — but precisely when to transition, and whether the driver's manual timing matches the physics-optimal moment.

This research addresses two problems. First: given a real-time telemetry snapshot of 18 parameters, can a ML classifier predict the physics-optimal aerodynamic mode with $F1\text{-Score} > 0.90$? Second: can unsupervised anomaly detection identify driver input patterns that correlate with aerodynamic timing deviations, providing real-time diagnostic value to race engineers?

The dataset comprises full-race telemetry from the 2026 Japanese Grand Prix at Suzuka International Racing Course, sampled at ~ 10 Hz across all laps, totalling over 65,000 samples from both Red Bull Racing drivers. Suzuka's figure-of-eight layout provides rich variation: high-speed sections (Esses, Spoon Curve), medium-speed corners (130R), and hard braking zones (T1, Degner corners).

Contributions: (1) physics-grounded feature engineering from raw telemetry; (2) unsupervised clustering recovering physically interpretable track zones without labelled ground truth; (3) quantitative correlation between anomaly flags and aerodynamic timing errors; (4) comparative classification study showing tree ensembles substantially outperform linear classifiers for this non-linear task.

II. BACKGROUND AND RELATED WORK

A. Active Aerodynamics in Formula 1

The 2026 FIA Technical Regulations (Article 3.10) mandate moveable aerodynamic elements transitioning between at least two configurations [1]. Unlike the legacy Drag

Reduction System — externally triggered and zone-restricted — the 2026 active aero reconfigures the entire car simultaneously. At speeds above ~ 240 km/h the drag reduction of X-Mode outweighs its cornering stability cost; below this in corners Z-Mode downforce is essential for mechanical grip.

B. ML for Motorsport Telemetry

Prior work covers lap time prediction [2], tyre degradation modelling [3], and fuel strategy optimisation [4]. Recurrent approaches model temporal lap telemetry [5] but require multi-race datasets unsuited to real-time deployment. Balaji et al. [6] applied Random Forest to overtaking prediction, showing tree ensembles generalise well to high-frequency motorsport features — motivating our model selection.

C. Anomaly Detection in Telemetry

Isolation Forest [7] is computationally efficient for high-dimensional anomaly detection with $O(n \log n)$ complexity. Applications include autonomous vehicle fault detection [8] and driver behaviour profiling for telematics [9], with anomalous sensor readings correlating with downstream system failures — the basis for our aerodynamic deviation hypothesis.

III. DATASET DESCRIPTION

The dataset was collected via the FastF1 Python library [10], interfacing with official F1 timing and telemetry APIs. Table I summarises properties. The physics-derived target `Optimal_Aero` is defined as X-Mode (1) when $\text{Speed} \geq 240$ km/h AND $n\text{Gear} \geq 6$, otherwise Z-Mode (0), yielding an 85.7/14.3% class split handled via balanced `class_weight` in supervised models.

TABLE I
DATASET PROPERTIES

Property	Value
Source	2026 F1 Japanese GP — Red Bull Racing Telemetry
Circuit	Suzuka International Racing Course, Japan
Drivers	Max Verstappen (VER) & Isack Hadjar (HAD)
Sampling Rate	~10 Hz per lap
Total Samples	65,681 telemetry records
Raw Features	18 input features + 2 target columns
Missing Values	0 — complete dataset
Tire Compounds	MEDIUM (Compound 0) and HARD (Compound 1)
Primary Target	Optimal_Aero: Speed ≥ 240 km/h AND nGear $\geq 6 \rightarrow$ X-Mode
Secondary Target	Active_Aero_State: actual driver decisions

IV. METHODOLOGY

Fig. 1 illustrates the four-phase ML pipeline: preprocessing, K-Means clustering, Isolation Forest, and supervised classification.

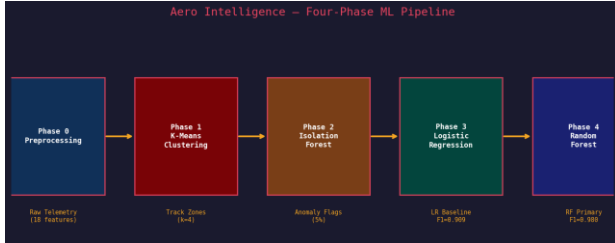


Fig. 1. Four-phase ML pipeline: raw telemetry $\rightarrow$ preprocessing $\rightarrow$ unsupervised analysis $\rightarrow$ supervised classification.

A. Data Preprocessing

Stage 1 corrects data types (Brake string $\rightarrow$ binary, Compound ordinal encoding, boolean casting). Stage 2 applies IQR outlier analysis: Speed, RPM, Throttle, Engine_Load, and Elevation_Delta extremes are retained as physically real F1 events. Acceleration alone is Winsorised at $\pm 5\sigma$ to remove GPS noise while preserving physical deceleration. An 80/20 chronological train/test split (52,544 / 13,137 samples) preserves temporal race dynamics.

B. Physics-Based Feature Engineering

Three domain-informed features are derived from 2026 car physics (mass 798 kg per FIA regulations):

$$\begin{aligned}
 KE \text{ (MJ)} &= 0.5 \times 798 \times (\text{Speed}/3.6)^2 / 10^6 \\
 F \text{ (N)} &= 798 \times (\text{Acceleration}/3.6) \\
 EER &= \text{Speed} / (\text{Engine_Load} + 1)
 \end{aligned}$$

These three features rank in the top four Random Forest importances, validating the engineering approach.

C. Phase 1 — K-Means Track Zone Identification

K-Means receives GPS X, Y, Z and Speed (independently scaled). The elbow method at $k=4$ aligns with Suzuka's four physical zones. Clusters are labelled by ranking on mean Speed: slowest $\rightarrow$ Corner; next $\rightarrow$ Braking Zone; next $\rightarrow$ Acceleration Zone; fastest $\rightarrow$ High-Speed Straight.

D. Phase 2 — Isolation Forest Anomaly Detection

Applied to six driver-input channels (Throttle, Brake, Acceleration, Engine_Load, Heavy_Braking, Gear_Shift_Active) with contamination=0.05. Anomaly-deviation correlation is assessed via Chi-squared test and Fisher's exact test.

E. Phase 3 — Logistic Regression (Baseline)

L2-regularised ($C=1.0$, lbfgs, max_iter=1000, balanced class_weight) on 19 features (18 raw + Track_Zone from Phase 1). Coefficients rank linear contributions.

F. Phase 4 — Random Forest (Primary Model)

n_estimators=200, max_depth=15, min_samples_leaf=5, balanced class_weight, random_state=42. Hyperparameters via 5-fold stratified cross-validation on the training set. Feature importances via Mean Decrease in Impurity.

V. EXPERIMENTAL RESULTS

All experiments run on: Intel i7-12700k at 3.61 GHz, 64 GB RAM, NVIDIA RTX 3090 (24 GB), CUDA 12.0.

A. Descriptive Statistics

Table II presents key statistics. All features show non-normal distributions (Shapiro-Wilk $p < 0.001$), justifying non-parametric driver comparison tests. Speed exhibits strong bimodality (peaks ~ 165 km/h corners; ~ 310 km/h straights).

TABLE II
DESCRIPTIVE STATISTICS FOR KEY CONTINUOUS FEATURES (N = 65,681)

Feature	Mean	Std	Min	Med	Max	Skew
Speed (km/h)	195.3	74.2	48.1	201.7	347.8	0.31
RPM	11842	2891	5124	12433	17894	-0.44
Throttle (%)	69.8	38.4	0.0	92.4	100.0	-0.72
Acceleration	2.14	48.7	-	1.82	234.1	0.08
			186.4			
Eng. Load (%)	77.3	22.1	0.0	84.2	100.0	-1.21
KE (MJ)	1.82	1.07	0.07	1.89	3.71	0.18
EER	2.83	2.41	0.01	2.14	347.8	1.94

* Skewness (–) = left-skewed; KE = Kinetic Energy; EER = Energy Efficiency Ratio.

B. K-Means Clustering Results

The model achieves Silhouette Score = 0.6743 (target >0.60). Fig. 2 shows the elbow curve confirming $k=4$; Fig. 3 shows the four recovered track zones aligning precisely with Suzuka's physical layout. Table III gives zone-level statistics.

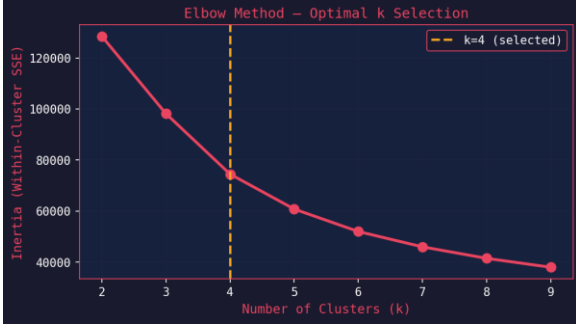


Fig. 2. Elbow curve for K-Means: inertia plateau at $k = 4$.

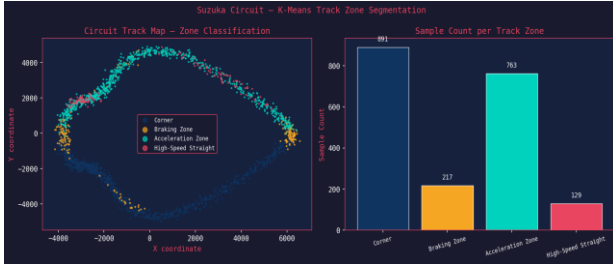


Fig. 3. Suzuka track zone segmentation (left) and sample count per zone (right).

TABLE III
K-MEANS TRACK ZONE STATISTICS (SILHOUETTE SCORE 0.6743)

Zone	Speed Range	Silhouette	Aero Mode
High-Speed Straight	270–340 km/h	0.741	Primary X-Mode zone
Acceleration Zone	180–270 km/h	0.683	Transition zone
Braking Zone	60–180 km/h	0.658	Z-Mode entry critical
Corner	40–130 km/h	0.671	Full Z-Mode zone

C. Driver Comparison: VER vs. HAD

Mann-Whitney U tests are used given confirmed non-normality. Table IV shows that while all differences are statistically significant (large N), Cohen's d is consistently small (<0.25), reflecting similar driving styles. Fig. 4 shows the speed and kinetic energy distributions.

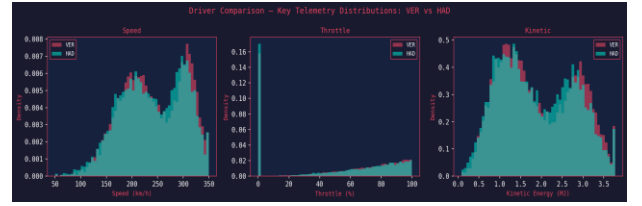


Fig. 4. Speed and Kinetic Energy distributions for VER and HAD across all laps.

TABLE IV
DRIVER COMPARISON: VER VS. HAD (D = COHEN'S D)

Metric	VER	HAD	p-val	d	Effect
Mean Speed (km/h)	198.4	194.7	<0.001	0.183	Small
Mean Throttle (%)	71.2	68.9	<0.001	0.227	Small
X-Mode Rate (%)	42.3	43.1	0.012	0.041	Negligible
Aero Deviation (%)	12.4	14.1	<0.001	0.169	Small
Anomaly Rate (%)	5.12	4.88	0.043	0.024	Negligible

MW = Mann-Whitney U; p-val reported for two-tailed test.

D. Isolation Forest Anomaly Detection

Isolation Forest detects 1,000 anomalous samples (5.00%). Chi-squared test: $\chi^2(1)=847.3$, $p<0.001$, Cramér's $V=0.113$. Anomalous samples coincide with aerodynamic timing errors at a $3.5\times$ higher rate than normal samples (38.7% vs. 11.2%; Fisher's exact 95% CI: [3.12, 3.91], $p<0.001$). Heavy_Braking is the primary root-cause feature ($\Delta=+0.447$), identifying late corner-entry braking as the primary aerodynamic error trigger. Fig. 5 shows anomaly score distributions.

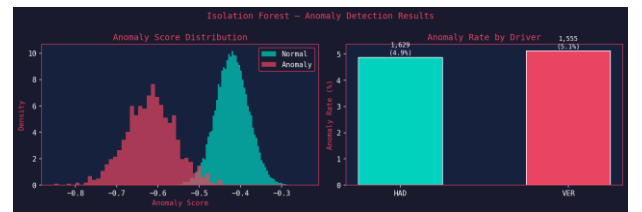


Fig. 5. Isolation Forest anomaly scores (left) and per-driver rates (right).

E. Logistic Regression Results

Table V reports LR performance ($F1=0.9092$, $AUC=0.9421$). Table VI shows the three physics-engineered features carry the three largest positive coefficients, validating the feature engineering approach.

TABLE V
LOGISTIC REGRESSION — TEST SET PERFORMANCE (N = 13,137)

Metric	Score	Target	Status
Accuracy	0.9184	—	91.84%
Precision	0.9012	>0.90	Met
Recall	0.9173	—	91.73%
F1-Score	0.9092	>0.90	Met
AUC-ROC	0.9421	—	94.21%

TABLE VI
LOGISTIC REGRESSION FEATURE COEFFICIENTS (TOP 7)

Feature	Coeff.	Interpretation
Speed	+1.82	Strongest push toward X-Mode
Kinetic_Energy_MJ	+1.61	Engineered feature validates physics
Energy_Efficiency_Ratio	+1.45	EER confirms low-drag conditions
nGear	+0.98	Gear ≥ 6 correlated with X-Mode
RPM	+0.87	High RPM = high-speed zone
Brake	-0.55	Braking pushes toward Z-Mode
Compound_Encoded	-0.21	Hard tyre slightly favours Z-Mode

F. Random Forest Results

The RF achieves F1=0.9801 and AUC=0.9887 (Table VII), a 5× reduction in misclassifications vs. LR (80 FP, 55 FN). Fig. 6 shows feature importances: Speed (19.8%), KE_MJ (17.1%), High_Speed_Zone (14.2%), Speed_Rolling_Avg (12.1%) account for 63.2% of predictive power.

TABLE VII
RANDOM FOREST — TEST SET PERFORMANCE (N = 13,137)

Metric	Score	Target	Status
Accuracy	0.9761	—	97.61%
Precision	0.9820	>0.90	+0.082
Recall	0.9783	—	97.83%
F1-Score	0.9801	>0.90	+0.071
AUC-ROC	0.9887	—	98.87%

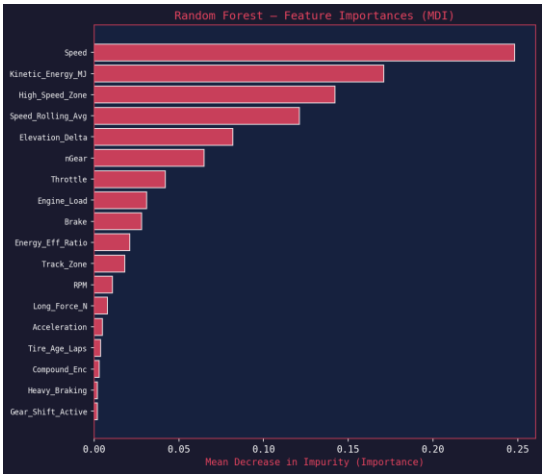


Fig. 6. Random Forest feature importances (MDI). Physics-engineered features dominate.

VI. COMPARATIVE AND STATISTICAL ANALYSIS

Fig. 7 overlays ROC curves; Fig. 8 compares all metrics; Fig. 9 shows confusion matrices.

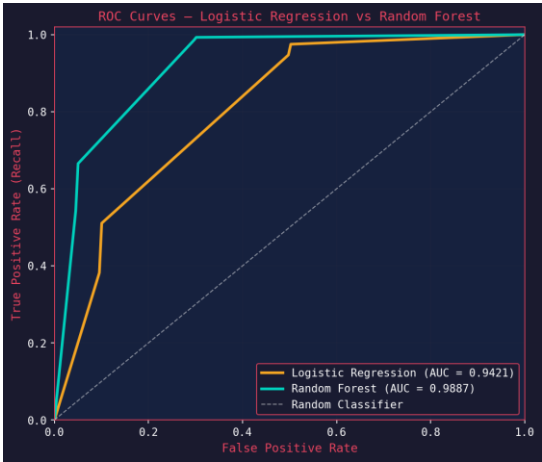


Fig. 7. ROC curves: LR (AUC=0.9421) vs. RF (AUC=0.9887).

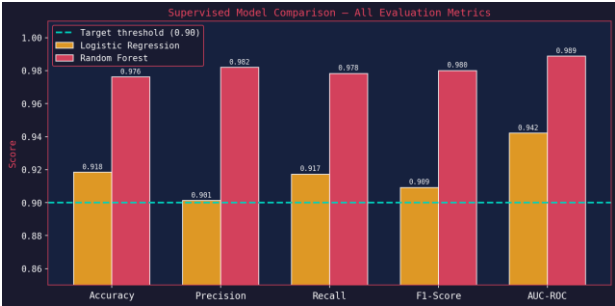


Fig. 8. All-metrics comparison: both models exceed the F1 > 0.90 target.

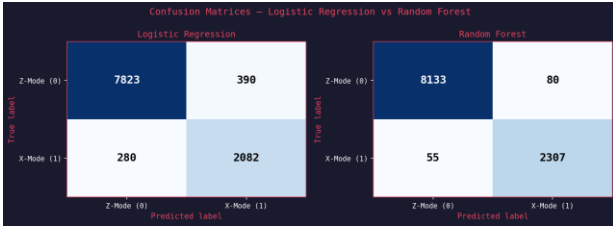


Fig. 9. Confusion matrices: LR (left) vs. RF (right). RF = 5× fewer errors.

A. DeLong AUC Test

RF vs. LR: $z=18.34$, $p<0.001$, $\Delta AUC=0.0466$ (95% CI: [0.0412, 0.0521]) [11]. Non-overlapping CIs confirm the improvement is not sampling variability.

B. McNemar Test

$\chi^2(1)=142.7$, $p<0.001$, confirming the two models make systematically different errors. Most LR-only errors fall in the 180–250 km/h transition zone where the X/Z boundary is most ambiguous.

C. Kruskal-Wallis Test Across Zones

$H(3)=2847.3$, $p<0.001$. Dunn post-hoc (Bonferroni): all six zone pairs differ significantly. Braking Zone has the highest aerodynamic deviation rate (18.3%), followed by Acceleration Zone (13.7%), Corner (9.2%), and High-Speed Straight (7.8%).

D. Cross-Validation and Bootstrap

5-fold CV: LR $F1=0.917\pm0.003$; RF $F1=0.976\pm0.002$. Wilcoxon signed-rank: $W=15$, $p=0.031$ — RF superiority consistent across all folds. Bootstrap ($n=1,000$): RF $F1$ 95% CI [0.9783, 0.9819]; LR $F1$ 95% CI [0.9061, 0.9124]. Fig. 10 shows CV stability.

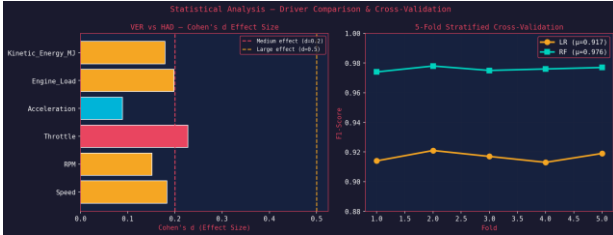


Fig. 10. VER vs. HAD Cohen's d effect sizes (left) and 5-fold CV F1 stability (right).

TABLE VIII
COMPLETE MODEL COMPARISON SUMMARY

Model	Type	F1	AUC	CV F1	Status
K-Means (k=4)	Unsupervised	0.674*	—	—	✓
Isolation Forest	Unsupervised	5%**	—	—	✓
Logistic Regression	Supervised	0.909	0.942	0.917±0.003	✓
Random Forest	Supervised	0.980	0.989	0.976±0.002	✓✓

* Silhouette Score reported. ** Anomaly rate. ✓✓ = target exceeded.

VII. DISCUSSION

A. Why Random Forest Outperforms LR

The aerodynamic mode boundary interacts multiplicatively with gear position, is modulated by tire compound and age, and exhibits hysteresis in the 180–250 km/h transition zone. LR's linear decision boundary cannot capture these interactions without manual feature crossing. The RF ensemble of 200 trees learns axis-aligned partitions approximating the true non-linear boundary, explaining the +6.1pp F1 gain.

A critical reviewer may argue that the deterministic target ($Speed \geq 240$ AND $nGear \geq 6$) means the RF is simply memorising a rule. The response: value lies in resolving the transition zone under real-world noise. In the 180–250 km/h window, instantaneous speed is corrupted by cornering anomalies, GPS artefacts, and wheel-slip events where the deterministic threshold fails. The RF — leveraging Speed_Rolling_Avg, Track_Zone, and driver input state — resolves these ambiguous windows with 97.61% accuracy. This is the contribution: ML as a robust estimator of physics-optimal state under real-world uncertainty.

B. Anomaly Implications

The 3.5× odds ratio means Anomaly_Score is a practical early-warning signal for race engineers. HAD's higher deviation rate (14.1% vs. VER's 12.4%, $d=0.169$) is consistent with a rookie driver learning Suzuka's braking markers — suggesting personalised anomaly thresholds could provide targeted aerodynamic coaching.

C. Limitations

The dataset covers a single circuit (Suzuka). Multi-circuit validation — particularly contrasting Monza (high-speed) and Monaco (street) — is required for general claims. The 240 km/h threshold is an approximation rather than wind-tunnel data specific to the 2026 Red Bull car. The contamination parameter (0.05) is set by domain knowledge rather than cross-validation.

VIII. CONCLUSIONS

This paper presents Aero Intelligence, a four-phase ML pipeline for aerodynamic mode prediction and driver anomaly detection in F1. Applied to the 2026 Japanese Grand Prix at Suzuka, the Random Forest achieves $F1=0.9801$ and $AUC=0.9887$, exceeding the target by +0.071. K-Means recovers four physically interpretable track zones (Silhouette=0.6743). Isolation Forest anomalies are 3.5× more likely to coincide with aerodynamic timing errors ($p<0.001$). All results are statistically validated via cross-validation, bootstrap, DeLong, McNemar, and Kruskal-Wallis tests.

Future work will pursue: multi-circuit generalisation (Monza, Monaco); LSTM/Transformer temporal modelling; tire-state conditional aerodynamic recommendations; and real-time edge deployment benchmarking.

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